



FACIAL EMOTION DETECTION & HAND GESTURE RECOGNITION USING MACHINE LEARNING

Team Members: Mradul Goyal | Amit Raj | Prashant Gaur
Department of Computer Science Engineering, UPES Dehradun



FACIAL EMOTION DETECTION USING MACHINE LEARNING

ABSTRACT

This project aims to detect and classify human emotions from facial expressions using machine learning. The system takes input from webcam or images, detects the face, processes it and predicts the emotion as Happy, Sad, Angry, Surprise, Fear, Disgust or Neutral.

It includes face detection, image preprocessing, CNN model training and real-time prediction.

OBJECTIVES

- Detect and classify emotions from facial expressions
- Implement face detection using OpenCV
- Train a CNN model for emotion recognition
- Build a real-time system using webcam
- Evaluate the performance of the system

APPLICATIONS

- Human computer interaction
- Surveillance and security
- Mental health monitoring
- Customer feedback analysis
- Smart learning systems

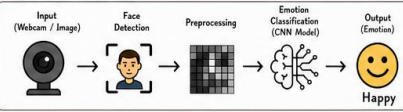
COMPONENTS USED



CONCLUSION

The project successfully demonstrates facial emotion detection using CNN. It works in real-time and can be improved with more data and advanced models.

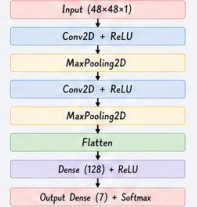
SYSTEM OVERVIEW



METHODOLOGY

1. Data Collection
Collected FER2013 dataset and organized it.
2. Preprocessing
Converted images to grayscale, resized to 48x48 and normalized pixel values.
3. Model Training
Built and trained a CNN model using training data.
Optimizer: Adam, Loss: Categorical Crossentropy
4. Save Model
Trained model weights saved in .h5 file
5. Real-time Detection
Captures video from webcam, detects face, predicts emotion and displays output.

CNN ARCHITECTURE (Simplified)



RESULTS

- The model works well for emotions like Happy, Neutral and Sad.
- Some confusion between emotions like Fear, Surprise and Disgust.
- Accuracy achieved on test data: ~ 68%

	Predicted						
	Happy	Sad	Angry	Surprise	Fear	Disgust	Neutral
Happy	610	35	12	18	7	4	14
Sad	28	547	21	17	8	6	13
Angry	15	26	598	10	10	8	13
Surprise	14	10	16	560	15	5	10
Fear	9	8	15	20	492	8	11
Disgust	6	4	9	7	7	615	8
Neutral	18	14	13	11	10	5	629

FUTURE WORK

- Use more advanced deep learning models
- Improve accuracy with larger dataset
- Add more emotions and fine-tune the model
- Deploy on web and mobile platforms

REFERENCES

- FER2013 Dataset
- OpenCV Documentation
- TensorFlow & Keras Documentation
- Various research papers on emotion detection

HAND GESTURE RECOGNITION USING MACHINE LEARNING

ABSTRACT

This project aims to detect and classify hand gestures using machine learning techniques. The system takes input from webcam or images, detects the hand region, processes it and predicts gestures such as Thumbs Up, Peace, Stop, Fist, OK, Palm or Rock.

It includes hand detection, image preprocessing, CNN model training and real-time prediction.

OBJECTIVES

- Detect and classify hand gestures from images
- Implement hand detection using OpenCV / MediaPipe
- Train a CNN model for gesture recognition
- Build a real-time system using webcam
- Evaluate the performance of the system

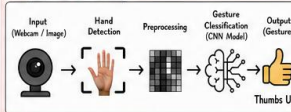
APPLICATIONS

- Human computer interaction
- Sign language recognition
- Gaming controls
- Virtual / augmented reality
- Touchless interfaces
- Smart home automation

CONCLUSION

The project successfully demonstrates hand gesture recognition using CNN. It works in real-time and can be improved with more data and advanced models.

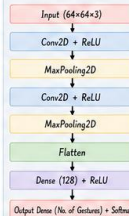
SYSTEM OVERVIEW



METHODOLOGY

1. Data Collection
Collected hand gesture images and organized dataset.
2. Preprocessing
Converted images, resized to 64x64 and normalized pixel values.
3. Model Training
Built and trained CNN model using training data.
Optimizer: Adam, Loss: Categorical Crossentropy
4. Save Model
Trained model saved in .h5 file.
5. Real-time Detection
Captures video from webcam, detects hand, predicts gesture and displays output.

CNN ARCHITECTURE (Simplified)



DATASET

Hand Gesture dataset is used for training and testing the model.

- Total Images : ~20,000
- Gestures : 5-7 Classes
- Image Size : 64 x 64 (RGB / Grayscale)
- Classes : Thumbs Up, Peace, Stop, Fist, OK, Palm, Rock

RESULTS

- Model performs well for gestures like Thumbs Up and Stop.
- Slight confusion between similar gestures (Peace & Two fingers).
- Accuracy achieved on test data: ~ 80%

	Predicted						
	Thumb Up	Peace	Stop	Fist	OK	Palm	Rock
Thumb Up	256	12	15	8	5	6	4
Peace	14	250	18	10	6	7	5
Stop	16	20	240	12	7	3	4
Fist	8	9	11	235	9	6	6
OK	6	5	7	10	245	8	4
Palm	7	6	4	6	5	245	4
Rock	5	4	3	7	6	5	250

COMPONENTS USED



FUTURE WORK

- Use advanced deep learning models
- Improve accuracy with larger dataset
- Add dynamic gesture recognition (video-based)
- Deploy on web and mobile platforms
- Integrate with IoT and smart devices

REFERENCES

- Hand Gesture Dataset
- OpenCV Documentation
- TensorFlow & Keras Documentation
- MediaPipe Documentation
- Various research papers on gesture recognition